

Mechanical Engineering Experiments (II)-Worksheet for Project Meeting

機械工程實驗(下)-專題討論工作單

Group information: (Fill your group number here No. 5)

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1. **DO NOT extend the space provided.** Learn to select representative points and leave detailed explanations to your slides. You may adjust the layout within the designated page if needed.
 2. Just leave blank for those slots you are not going to fill.
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Meeting date:

Credit:

Group leader's signature:

Feedback from Lecturer:

Lecturer signature:

Part (a) What have you done

1. Chose our approach for this project

- (a) Brainstormed all possible solutions
- (b) Assessed all potential solutions and finalized the design selection

2. Decided the key specs for our design

- (a) Ball picking mechanism
- (b) Estimated whole drone weight required for the ball picking payload
- (c) Find the minimum requirement of every part of the system

3. Picked the parts for our drone (below are parts and their reasoning)

(a) Flight Controller:

We selected the **F405 AIO 40A** flight controller to have ease of integration (All In One).

- Integrated a **40A ESC** onto a single **25.5 x 25.5 mm** PCB to achieve a light weight.
- Allowing for a higher mechanical payload capacity. The F405 was specifically chosen over - **40A current rating** provides the thermal headroom to drive our high-torque motor.

(b) Motor and Propeller:

We selected the **EMAX RS2205** brushless motor paired with **5045 tri-blade propellers**.

- Integrated Active Cooling: The cooling fins that reduce operational temperatures by **30%**.
- Optimal Thrust-to-Weight Ratio: Generating over **1,024g of thrust** on a 4S LiPo with 5045 propellers, a four-motor configuration provides a total thrust exceeding **4kg**.

(c) Battery:

We selected a **14.8V (4S) 2700mAh 35C Li-Po** (232g) to provide a high-voltage platform.

It has high energy density with optimal voltage (14.8V), the 4S configuration delivers the necessary RPM and torque.

- Balanced Discharge (35C): Provides a reliable 94.5A output.

- Voltage selection: Running a 4S system reduces current draw compared to 3S for the same power output, minimizing heat buildup in the ESCs and motors during sustained operation.

4. Material selection for the frame with experiments

We conducted a comparative study between MDF (Medium-Density Fiberboard) and 3D-printed PLA to identify the most rigid material for the drone arms. By combining digital simulations with physical destructive testing, we analyzed how different materials handle the high-torque demands of our propulsion system.

-Digital FEA Simulation: Utilized 3D modeling to identify critical stress concentration zones (max stress: 33.78 MPa) near the arm-to-body transition.

-Physical Prototyping: Fabricated test arms using laser-cut MDF (3mm/5mm) and 3D-printed PLA to compare weight-to-stiffness ratios.

-Destructive Testing: Applied incremental loads to the motor mounts in a cantilever setup, measuring the specific displacement until failure.

-Experimental Results: MDF exhibited significantly higher rigidity (steeper slope on the Force/Displacement graph) and a higher fracture point (up to 3425 gw) compared to PLA (2745 gw).

Final Selection: Choose MDF for the frame due to its superior Young's Modulus, ensuring the arms remain rigid during high-thrust maneuvers to prevent flight instability.



Figure 1: Clamping system of the experiment

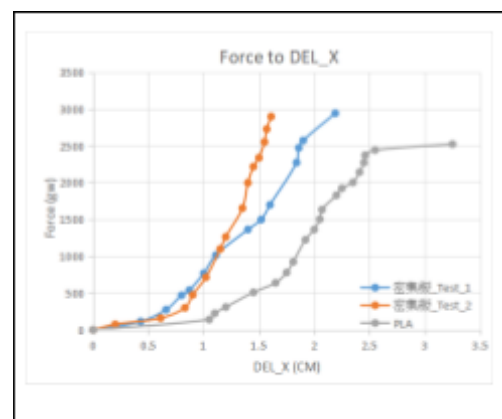


Figure 2: Force vs DEL_X plot

Part (b) What barriers you encounter:

1. Battery voltage sag issue: Our 4 motors can pull ~120A at full throttle, which exceeds the battery's maximum safe discharge limit of 94.5A ($2.7\text{Ah} \times 35\text{C}$).

The Risk: Punching the throttle to lift the payload will cause a massive voltage drop (sag), potentially triggering the flight controller's low-voltage cutoff and crashing the drone.

Potential Solution: I will configure a hard "Motor Output Limit" (capped around 80%) in the flight controller software. This creates an artificial ceiling to prevent peak current spikes, trading unnecessary top speed for guaranteed electrical stability.

2. Propeller Downwash Issue

Propeller downwash may blow the tennis ball away, making positioning and gripping difficult. We are concerned that when the drone hovers at a specific distance above the platform, the pressure generated by the downwash exceeds the static friction between the tennis ball and the platform surface. We came up with a few preliminary thoughts on the experimental design. First, we have to figure out how the friction works between the ball and the platform. We can vary the motor RPM and the hovering altitude to figure out how the balls would be moved. Whether or not the ball moves under downwash will significantly influence the final gripper design.

3. Rule compliance for purchased propellers: The propeller data from the manufacturer is rather limited, we need to come up with some experiments to justify the choosing of this current propeller set.
4. We still lack the knowledge of integrating our servo motor controlling our ball gripper into the flight controller board.
5. Regulatory procurement restrictions: Current prohibitions regarding Chinese-manufactured drone components have significantly impeded the acquisition phase, causing a substantial stall in our hardware procurement timeline.
6. Servo-to-FC integration: Current technical knowledge regarding the hardware-level interfacing of the ball-gripper servo motor with the flight controller board remains insufficient.
7. Servos are controlled via Pulse Width Modulation, so we have to understand how a pulse corresponds to the servo's angle first. Additionally, high-torque servos for grippers can draw significant current. If the flight controller's internal power regulator is overloaded, the board might reboot mid-flight.

Figure 3: RS2205 performance specifications supporting the analysis of excessive current draw

Motor type	The voltage (V)	Paddle size	current (A)	thrust (G)	power (W)	efficiency (G/W)	speed (RPM)
RS2205 -2600KV	12	HQ4045 BN	1	55	12.00	4.58	8030
			3	131	36.00	3.64	12350
			5	196	60.00	3.27	14910
			7	256	84.00	3.05	16940
			9.1	308	109.20	2.82	18700
			11	362	132.00	2.74	20200
			13.1	420	157.20	2.67	21570
			15.1	461	181.20	2.54	22770
			17.2	510	206.40	2.47	23170
			19.5	567	234.00	2.42	24820
	16	HQ4045 BN	1	65	16.00	4.06	8980
			3	153	48.00	3.19	13380
			5	232	80.00	2.90	16240
			7	293	112.00	2.62	18130
			9	352	144.00	2.44	19930
			11	420	176.00	2.39	21410
			13.1	466	209.60	2.22	22900
			15.2	521	243.20	2.14	24190
			17.1	574	273.60	2.10	25340
			19.1	620	305.60	2.03	26390
			21	670	336.00	1.99	27260
			23.1	732	369.60	1.98	28100
			25	763	400.00	1.91	29050

Part (c) What is your next step:

1. Hardware Procurement & Core Assembly

Prototype Build: Complete the power distribution wiring and basic FC configuration to assemble a minimum viable drone.

Initial Flight Testing: Conduct the first tethered or low-altitude hover test to verify stability, vibration levels, and thrust-to-weight performance.

2. Gripper Mechanism & Structure Design

Design Iteration: Draft the first-version gripper. We can focus on the L-Shaped 3-Sided Constraint to shield the tennis ball from wind interference.

Passive Geometry: Refine the Passive Gravity Funnel design, prioritizing a semi-conical shape that guides the ball into position without needing heavy servos or complex actuators.

Spatial Mapping: Finalize the drone's frame layout and configuration diagrams to identify the exact mounting points and available volume for the gripper assembly.

3. Material Analysis & Optimization

Property Comparison: Continue evaluating various materials (e.g., standard PLA vs.MDF) and wall thicknesses to balance structural rigidity with the strict weight requirements of the drone.

Weight Budgeting: Ensure the proposed gripper designs remain lightweight to prevent overloading the motors and reducing flight time.

4. Experimental Validation (Downwash & Aerodynamics)

Propeller Downwash Test: Once motors are mounted, conduct controlled experiments to measure how propeller downwash affects tennis ball stability at different heights.

Performance Tuning: If tests show high current draw or excessive turbulence, evaluate switching to a lower-load propeller pitch or diameter.

Research: Perform a deep dive into lift and propeller analysis methods to better predict flight behavior under the payload of the ball-fetching mechanism.



Figure 5: first prototype of the gripper design

List your reference in this page:

For website references, please provide the *source*, *page title*, and *URL*. For journal paper, apply the APA format. Examples are show below, [1] for a website, [2] for a formal journal paper:

[1] National Instrument, *what is DAQ*, <https://www.ni.com/zh-tw/shop/data-acquisition.html>

[2] Box, G. E. (1976). Science and statistics. *Journal of the American Statistical Association*, 71(356), 791-799.

[3] Emaxmodel, *emax rs2205 racespec motor cooling series*,
<https://emaxmodel.com/products/emax-rs2205-racespec-motor-cooling-series>

[4] Tyto Robotics, *how much weight can a drone carry*,
<https://www.tytorobotics.com/blogs/articles/how-much-weight-can-a-drone-carry>

[5] Omni Calculator, *drone motor*, <https://www.omnicalculator.com/other/drone-motor>

[6s] Zbotic, *drone battery voltage sag why it happens and how to fix it*,
<https://zbotic.in/drone-battery-voltage-sag-why-it-happens-and-how-to-fix-it/>

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